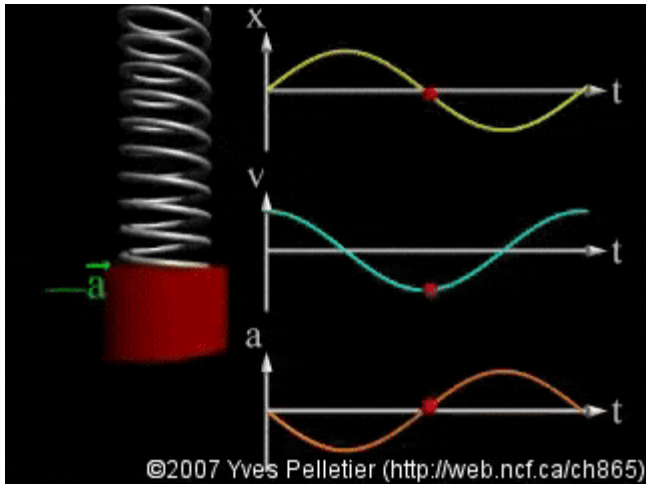
PHYC10009 Foundations of Physics – lecture 12



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(Please switch your mobile phone to silent)

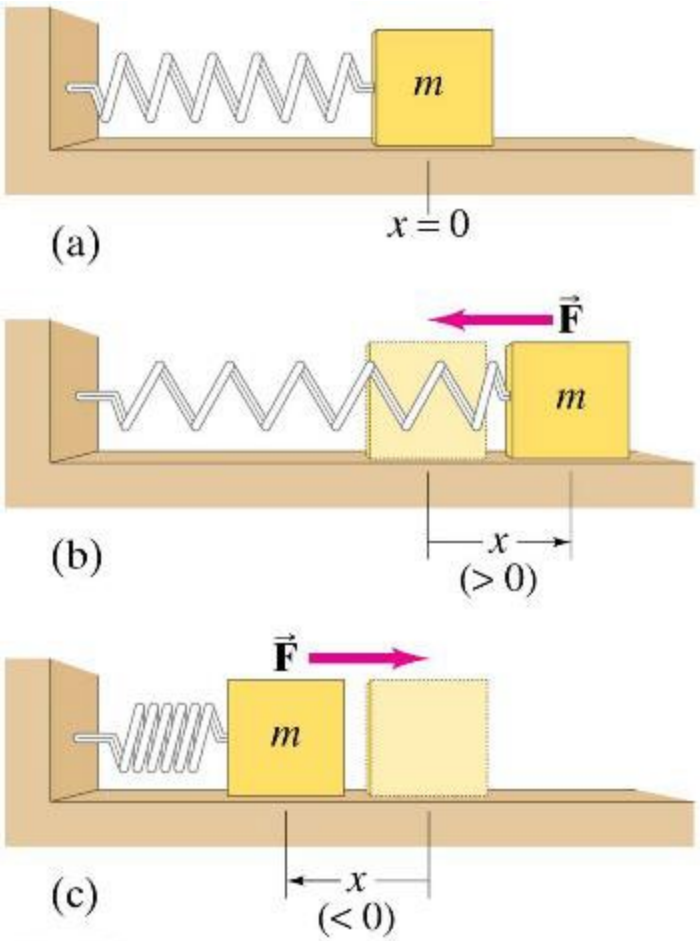
Simple Harmonic Motion G11-1

Hooks Law for Springs: $F = -k\Delta x$

Mean position: $\Delta x = 0, F = 0$

Positive displacement: $\Delta x > 0\ , F < 0$

Negative displacement: $\Delta x < 0\ , F > 0$



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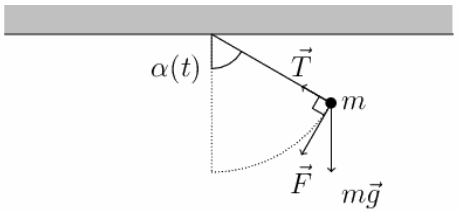
Simple Harmonic Motion: mass and spring



Oscillations everywhere!

Mass on spring, child on swing,

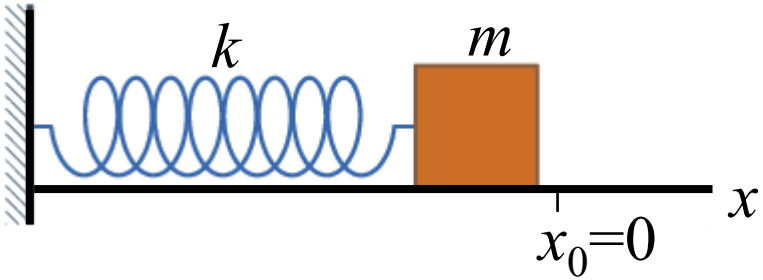
pulsar, planets in orbit, atomic clocks, …

k = spring constant

$$F = -kx$$

Simple Harmonic Motion occurs in all systems where the *<u>restoring force is proportional to the displacement</u>*

Any repetitive motion is harmonic motion



What are the *<u>units</u>* of the argument of the cosine function?

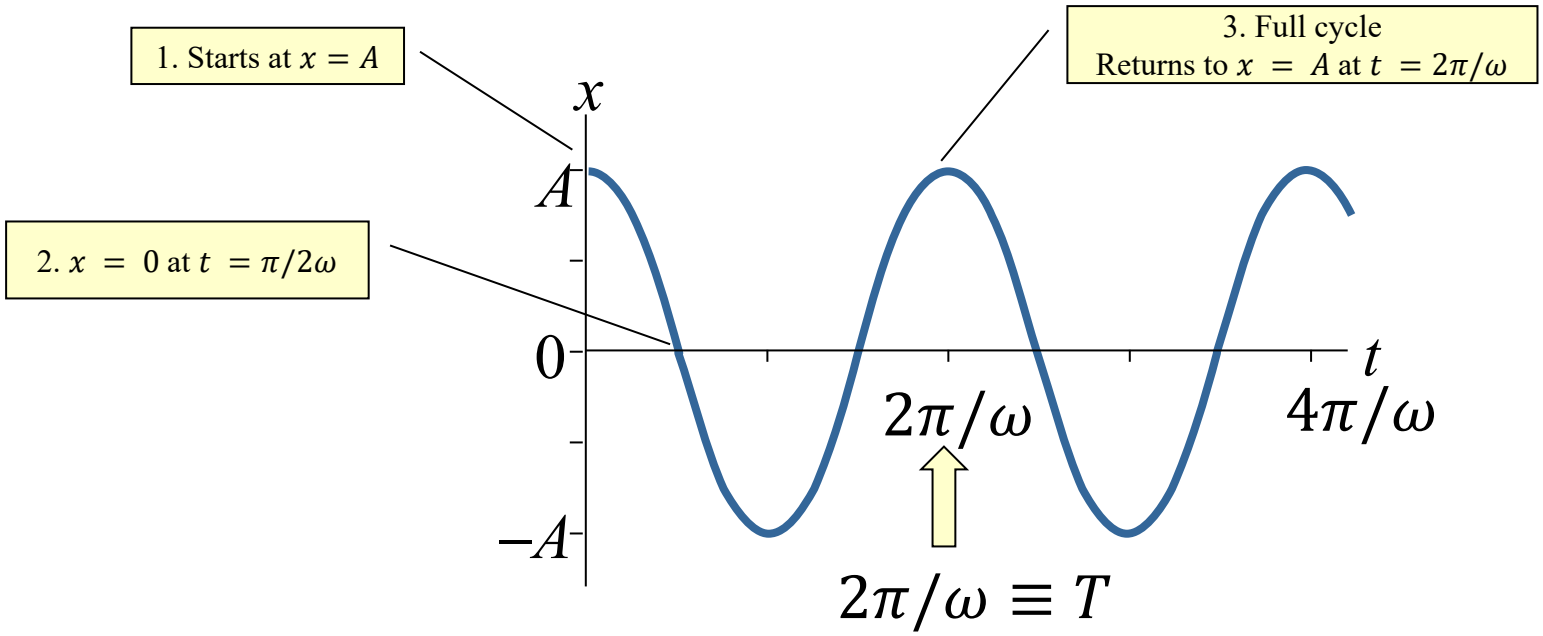
$$x(t) = A\cos(\omega t)$$

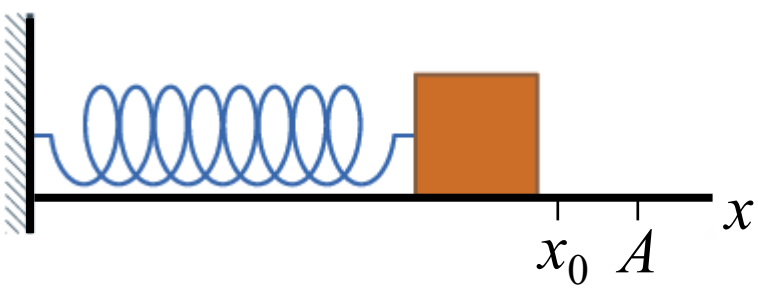
Describe motion using a *sinusoidal function*, e.g. cosine

Frequency ω in radians/second, time t in seconds

Graphically

$$x(t) = A\cos(\omega t)$$





Angular frequency

We have used *angular* frequency ω, radians per second

Note units: radians per second, written $\boxed{s^{-1}}$

Oscillation frequency *f* in *Hertz* (cycles per second)

$$f = \frac{\omega}{2\pi} \quad [\mathrm{Hz}] \qquad \omega = 2\pi f \quad [\mathrm{s}^{-1}]$$

Period is $T = \frac{1}{f} = \frac{2\pi}{\omega}$

Phase = angle = argument of cos(...) or sin(...)

Angular *frequency* is rate of change of this "angle" (phase)

Position of the object in SHM is then

$$x(t) = A\cos(\omega t) \equiv A\cos\left(\frac{2\pi}{T}t\right)$$

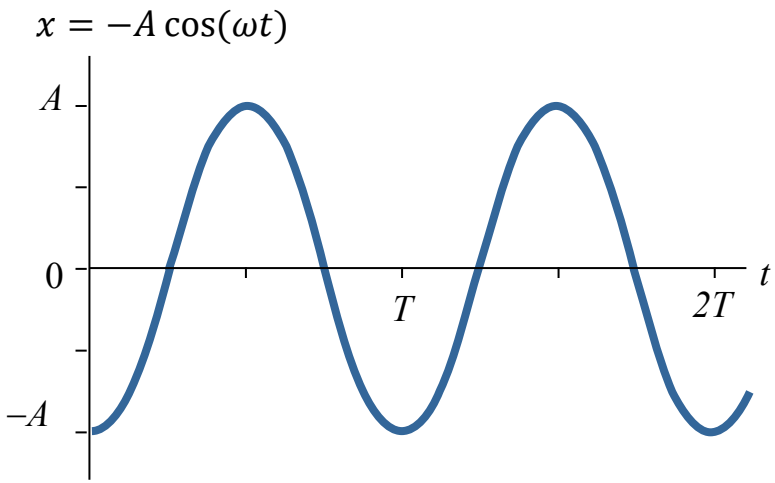
Example exam-style question

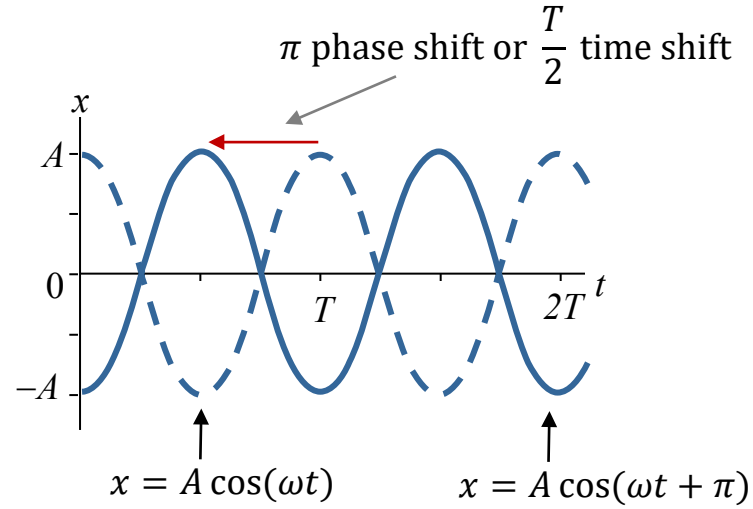
Consider the motion of a simple harmonic oscillator described by

$$x(t) = -A\cos(\omega t)$$

1. Sketch a displacement-time graph for the oscillator and label the graph with terms in the equation.
2. At what position is the particle at time $t = 0$?
3. Determine the phase constant ϕ_0 in the default equation of motion $x(t) = A\cos(\omega t + \phi_0)$ such that both describe the same motion.

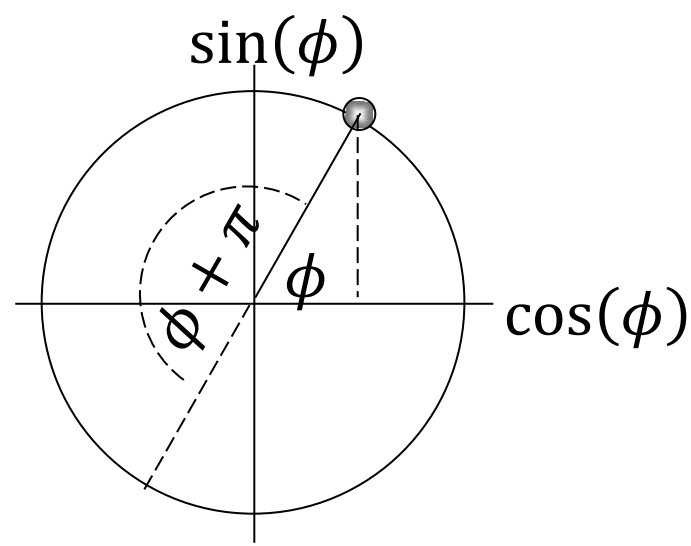
Example exam-style question





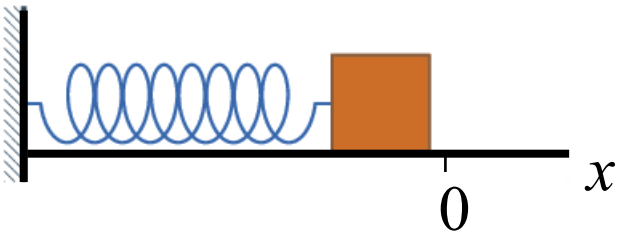
$$x(t) = -A\cos(\omega t) = A\cos(\omega t + \phi_0)$$

$$A\cos(\omega t + \pi) = -A\cos(\omega t)$$



Velocity

SHM $x(t) = A\cos(\omega t)$

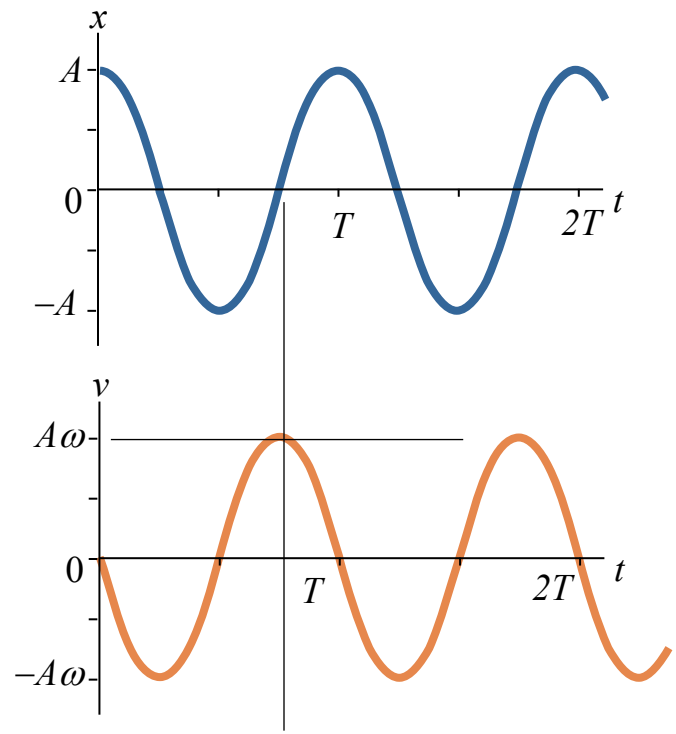


Velocity from derivative

Proportional to A

$$v_x(t) = \frac{dx}{dt} = -A\omega\sin(\omega t)$$

Proportional to ω



What is maximum velocity?

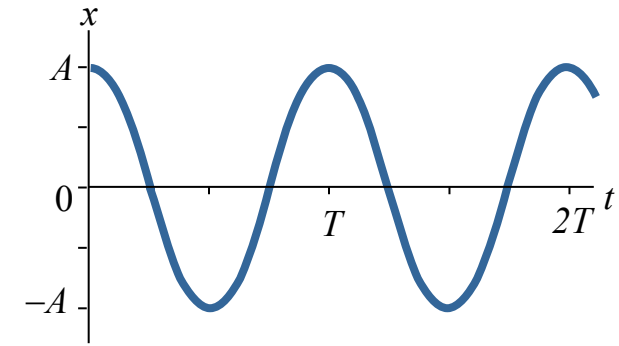
Where in cycle does maximum velocity occur?

Note $v = 0$ at $x = \pm A$

Acceleration

Displacement

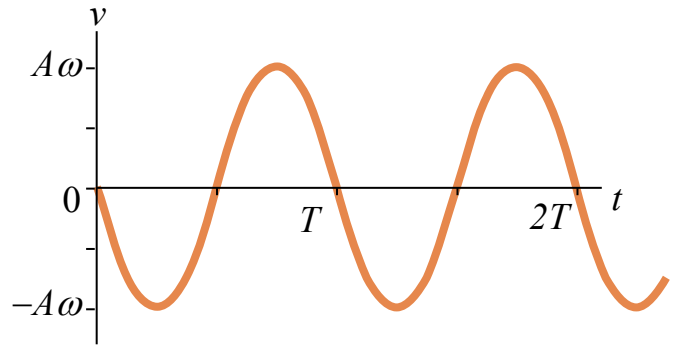
$$x(t) = A\cos(\omega t)$$



Velocity

Note the phase shift

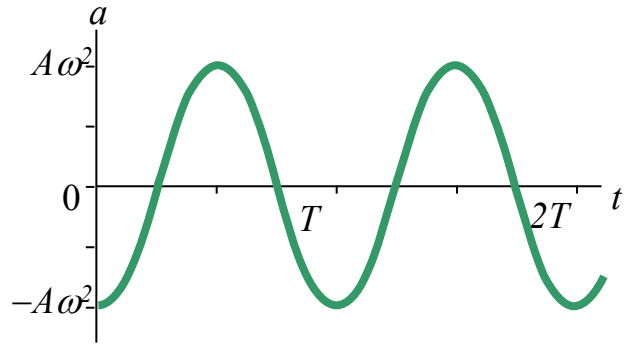
$$v(t) = \frac{dx}{dt} = -A\omega\sin(\omega t)$$



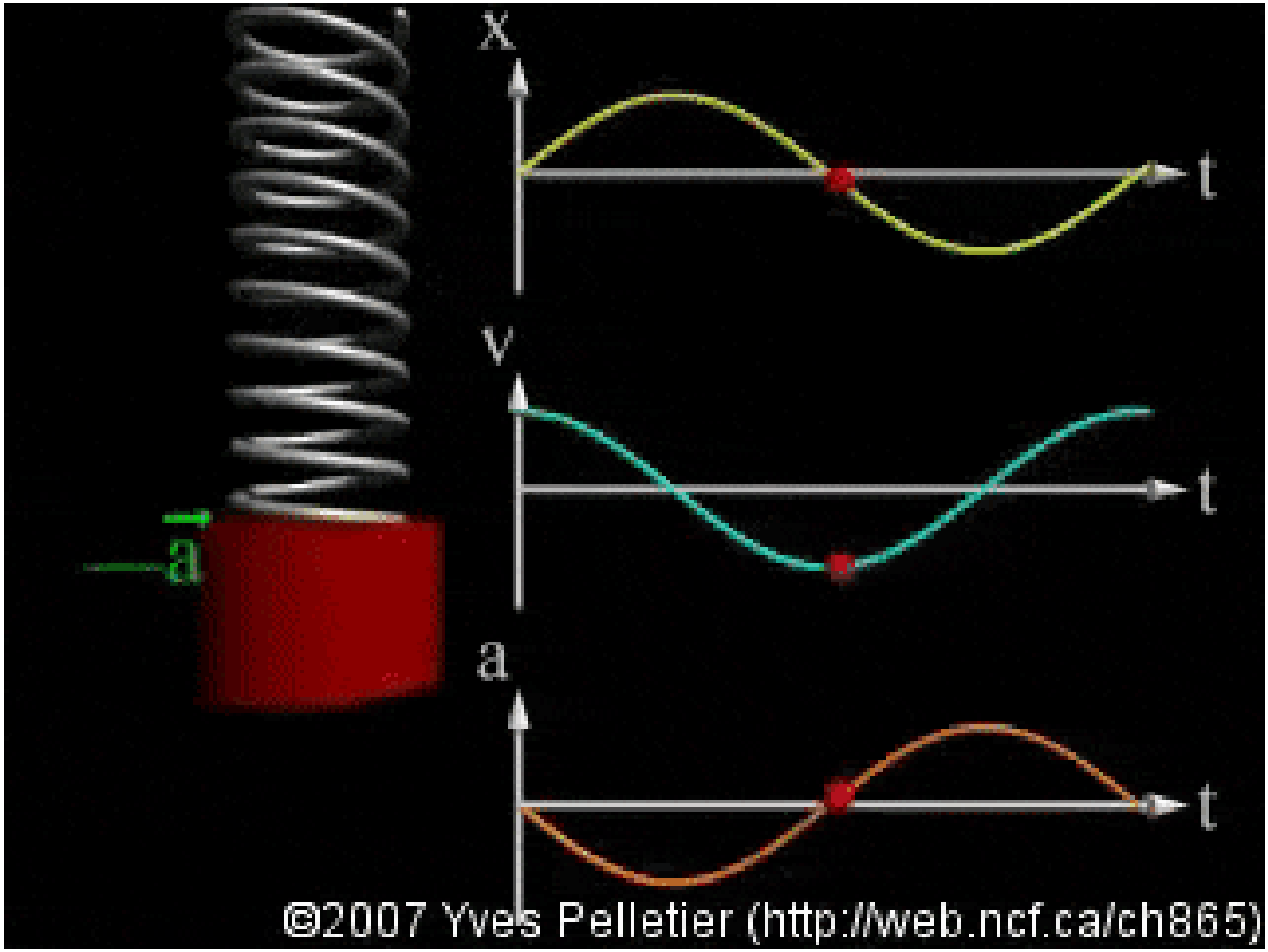
Acceleration

Note the phase shift

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -A\omega^2\cos(\omega t)$$



Watch the Animation



The dynamics of SHM G11.3

SHM and Hooke's Law

$$x(t) = A\cos(\omega t)$$
$$v_x(t) = -A\omega\sin(\omega t)$$
$$a(t) = -A\omega^2\cos(\omega t)$$

Acceleration from second derivative

But $x(t) = A\cos(\omega t)$ so $\quad a(t) = -\omega^2 x(t)$

Force $F = ma \Longrightarrow \quad F = -m\omega^2 x$

That is, $\boxed{F = -kx}$ where $k = m\omega^2$

$$\omega = \sqrt{\frac{k}{m}}$$

This is the hallmark of SHM!

"In SHM the <u>restoring</u> force is proportional to the displacement"

i.e. opposite in sign

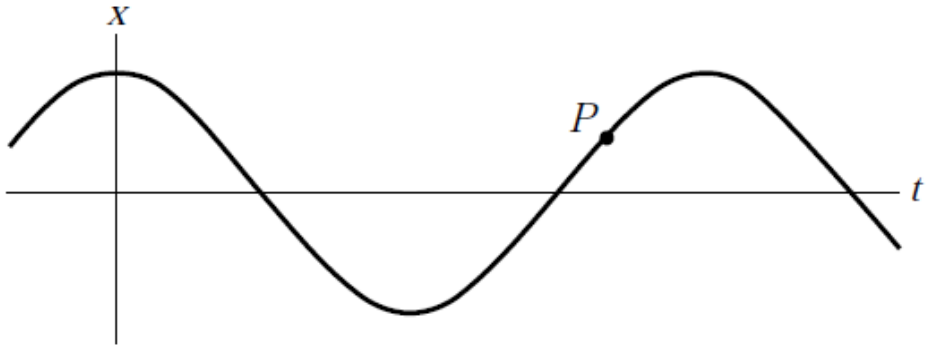
i.e. acceleration

D Md-1 Mass on a spring

Recall: $\omega = 2\,\pi\,f = \frac{2\pi}{T} \Rightarrow \mathrm{T} = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$

Concept test

A mass attached to a spring oscillates back and forth as indicated in the position vs. time plot below. When the mass is at point *P*, what can you say about the particle's position and velocity?

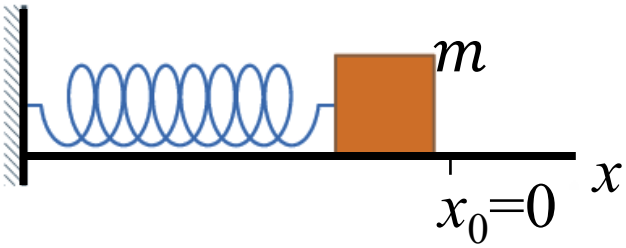


a. The position and velocity are both positive
b. The position and velocity are both negative
c. The position is positive and the velocity is zero
d. The position is negative and the velocity is zero
e. The position is positive and the velocity is negative

Answer at https://PollEv.com/martinsevior
Or on your Mobile Phone

Example exam-style question

Consider the motion of a mass m on a spring (see figure), where the mass moves on a frictionless surface, and the spring force is given by $F = -kx$. The at-rest position is $x = x_0$=0. The mass is displaced to some maximum distance x_m and released at time $t = 0$.



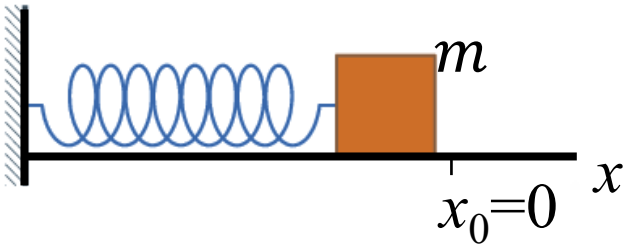
a) Sketch the displacement x as a function of time, for three oscillations.

b) Write an equation for the displacement x as a function of time, if the period of oscillation is 1 second.

c) Sketch the velocity v as a function of time.

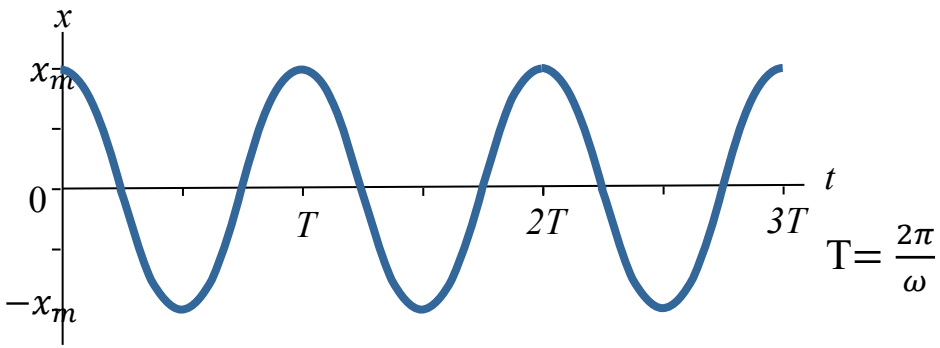
d) Using your answer to part b), derive an equation for the velocity.

Solution:



Displaced to max distance $x = x_m$ at $t = 0$

(a)



(b) General formula for SHM $x(t) = Acos(\omega t + \phi_0)$

Period T = 1 sec = $\frac{2\pi}{\omega} \Rightarrow \omega = 2\pi$

Amplitude (A) = x_m

When $t = 0, x(0) = x_m \cos(\phi_0) \Rightarrow \phi_0 = 0$

So: $x(t) = x_m \cos(2\pi t)$

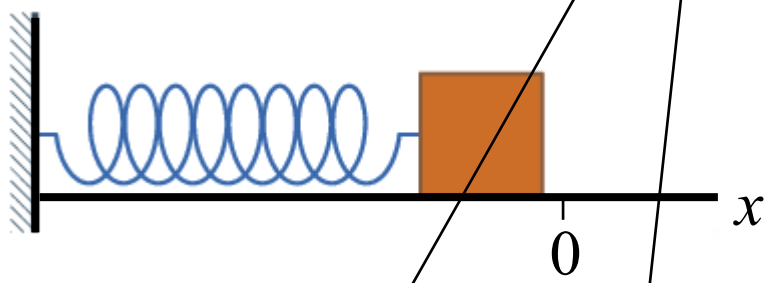
(c) $$v(t) = \frac{dx}{dt} = -2\pi \sin(2\pi t)$$

Energy in SHM G11.2

$$x(t) = A\cos(\omega t)$$
$$v_x(t) = -A\omega\sin(\omega t)$$

SHM: interplay between *kinetic* and *potential* energy

<u>Total</u> mechanical energy E is constant



Mechanical energy $E = K + U$

For a spring:

Potential energy $U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2(\omega t)$

Kinetic energy $K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2A^2\sin^2(\omega t)$

Total energy $E = U + K = \frac{1}{2}kA^2\cos^2(\omega t) + \frac{1}{2}m\omega^2A^2\sin^2(\omega t)$

$\omega = \sqrt{\frac{k}{m}} \Rightarrow \omega^2 = \frac{k}{m}$ So $E = \frac{1}{2}kA^2\cos^2(\omega t) + \frac{1}{2}m\frac{k}{m}A^2\sin^2(\omega t)$

$$E = \frac{1}{2}kA^2(\cos^2(\omega t) + \sin^2(\omega t)) = \frac{1}{2}kA^2$$

So Total Energy is constant

Energy in SHM G11.2

$$x(t) = A\cos(\omega t)$$
$$v_x(t) = -A\omega\sin(\omega t)$$

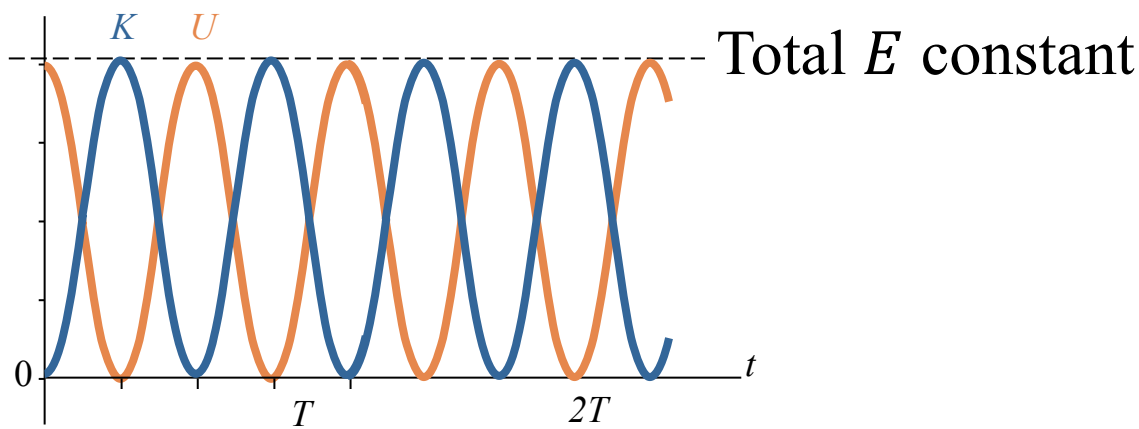
Total energy $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$

Kinetic zero, potential max: $E = U = \frac{1}{2}kx_{max}^2 = \frac{1}{2}kA^2$

Potential zero, kinetic max: $E = K = \frac{1}{2}mv_{max}^2 = \frac{1}{2}m(A\omega)^2$

Since $\omega = \sqrt{\frac{k}{m}}$ $\quad \frac{1}{2}kA^2 = \frac{1}{2}m(A\omega)^2$

Energy continuously transforms between Kinetic and Potential. The Total energy remains constant at all times.

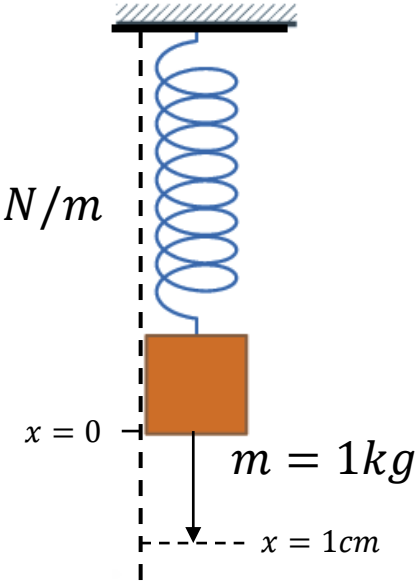


Sample question

A mass of 1.0 kg is fixed to a spring with spring constant $k = 100\ \mathrm{N/m}$. The spring is stretched by 1.0 cm from its resting position and then released.

a) Calculate the oscillation frequency.
b) Calculate the maximum potential energy and the maximum kinetic energy.
c) Calculate the time after release when the kinetic energy is maximal.

Solution



(a) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{1}} = 10\,\frac{rads}{sec}$

$\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi} = \frac{10}{2\pi} = 1.6\ Hz$

(b) Potential Energy, $U = \frac{1}{2}kx^2$

U is maximum when x is max. so

$$U_{max} = \frac{1}{2} \times 100 \times (0.01)^2$$

$$\Rightarrow U_{max} = \frac{1}{2} \times 100 \times 10^{-4} = 5 \times 10^{-3} J$$

$$x(t) = Acos(\omega t) = \ 0.01\cos(10t)$$

$$\Rightarrow v(t) = \frac{dx}{dt} = -0.01 \times 10\sin(10t) = -0.1\sin(10t)$$

$K = \frac{1}{2}mv^2$ is max, ie when v is max, which is $v = 0.1$

$$K_{max} = \frac{1}{2} \times m \times v^2 = 0.5 \times (0.1)^2 = 5 \times 10^{-3} J$$

(c) K is max when $\sin(10t) = 1 \Rightarrow 10t = Sin^{-1}(1) = \frac{\pi}{2}$

$$t = \frac{\pi}{20} = 0.16\ sec$$

Summary of Lecture 12

Generic SHM: $x(t) = A\cos(\omega t) = A\cos\left(\frac{2\pi}{T}t\right) = A\cos(\phi)$

- *Angular frequency* $\omega \quad \left[\frac{\text{rad}}{\text{s}}\right]$ $\qquad \omega = \sqrt{\frac{k}{m}}$
- *Amplitude* A [units depend on application!]
- *Frequency* $f = \frac{\omega}{2\pi}$ [Hz]
- *Period* $T = \frac{1}{f} = \frac{2\pi}{\omega}$ [s]
- *Phase* $\phi = \omega t + \phi_0$
- In SHM the <u>restoring</u> force is proportional to the displacement

$$a_x(t) = \frac{dv}{dt} = \frac{d^2x}{dt^2} = -A\omega^2\cos(\omega t) = -\omega^2\, x(t)$$

Common mathematical difficulties

Unfamiliarity with radians

The argument of an oscillatory function needs to be an *angle*

no *physical* units

E.g. $\sin(t)$ is not sensible (if t is time in seconds)

Know simple trigonometric identities

$\sin(-x) = -\sin(x)$ or $\sin(x + \pi/2) = \cos(x)$ and so on

sine and cosine are not different functions! cos(x) is a sinusoidal function

Be sure you can draw $\sin(\omega t + \pi/2)$ as opposed to $\sin(\omega t - \pi/2)$

Sine and cosine functions familiar – but why are these trigonometric functions relevant to *oscillations?*

The concept of *phase*